

comfort of a self-driving vehicle?

To answer your question, I think the bigger adoption will take place in vehicles which mostly ply on national highways and talking about autonomous driving in cities will remain a far-fetched idea.

Talking about autonomous vehicle adoption in India vs Europe and USA, what are your views?

I think the difference between India and the USA or Europe at the primary level is, when you are in the latter countries, you are basically looking to eliminate drivers because you are paying them incredibly high salaries. In countries that fall in Europe and in the USA, there are extremely strict rules for commercial drivers. They are entitled to specific amounts of driving every day, and that driving must be accompanied with breaks specified in their laws. These laws make commercial drivers awfully expensive in those countries, whereas no one follows any such rules in India. Commercial drivers keep driving endlessly on the Indian roads.

It is extremely badly regulated in India and the commercial drivers here cost almost a fraction of the cost in Europe and America. In addition, they drive much longer than drivers of these countries. That cost is one that is playing a major factor in differentiating the adoption of autonomous driven tech in such countries.

What about personal cars?

The same cost also plays a role in adoption of the tech in terms of driving a personal vehicle in India. For instance, if I get drunk, I can enjoy the privilege of being driven home by a driver at exceptionally low cost. So, why would I want to spend a lot more on autonomous driving tech?

The other thing is that India has never been safety conscious. We are talking about auto driving tech, but can you tell me how many people use helmets on motorbikes and seatbelts in cars? And that is when these are the most basic safety requirements for vehicles plying on Indian roads. Can you also tell me how many people know the safety risks associated with seating six people in a car certified for five people? No one tests vehicles for such scenarios. We, in India, are not safety conscious essentially!

I think that bigger adoption of autonomous tech will come in the commercial sector by the transport companies managing big fleets. For the

Problems around autonomous driving tech that need focus

1. The traffic conditions in India are not similar to anywhere else in the world. We need focused research around the same.
2. Infrastructure to enable autonomous driving needs to be put up. For example, driving lanes need to be marked clearly on roads.
3. Traffic laws need to be implemented seriously
4. Investing firms need to get serious about long-term investments in this deep tech.

Where are the opportunities

1. There is plenty of room for startups to research and come up with India-specific autonomous driving software.
2. Hardware used for enabling autonomous driving can be connected to drones, robots, and Industry 4.0. Enabling a hardware manufacturing ecosystem will mean growth for all these sectors.
3. India has the opportunity to lead and make this tech for a lot of countries in Asia and Africa.
4. If successful in India, the software and hardware can be licensed and outsourced to any other part in the world.

private sector, the tech might remain a luxury that only the rich would want to test given the road conditions on the streets.

Can the government do something about it?

Let me start by saying that owning a basic car, in many parts of India, is still seen as a matter of pride and less of a necessity. Therefore, the complete business model gets into chaos and the government shows less interest because self-driving technology is not a major immediate national problem. In terms of commercial long-haul trucks, there are always at least two drivers on board. In terms of cab driving, the same is becoming an exceptionally large job sector in India. Then a lot of car owners use full-time or part-time drivers for private cars in most of the urban locations.

Autonomous driving tech might cause an immense loss of jobs, and I do not think the government will want to do that. As of now, the scale of benefits is lighter than the scale of losses. It is also not fair to leave everything to the government. The user attitude must be different as well.

The current user attitude in India is not allowing the commercial model of autonomous driving tech to come up in the country. I think the difference in commercial interests is the real problem in India.

The government is trying to increase adoption of EVs, would EVs packed with autonomous driving tech appeal more to people?

When you talk about EVs and

autonomous driving tech, I agree that it is easier to enable an EV with the tech than enabling an ICE vehicle with auto driving tech, and yes there is more than enough proof that this tech is slowly becoming a USP in electric vehicles.

However, when we talk about the present levels of autonomy in electric vehicles, the advantages of EVs cannot go beyond L1, L2, or at the max L3 level of autonomous driving tech. The question to ask is, does the advantage of EVs matter in terms of L5 level? I do not think it would matter paying a little extra for a little piece of hardware and autonomous driving software when a vehicle already has tens of lakhs (millions of rupees) worth of lidar sensors.

Essentially, that is more than the cost of the vehicle itself, and in the long run the software licenses may cost more than the hardware. I do not think the type of engine, whether petrol or electric, will matter in terms of adoption of autonomous vehicles that offer L5 capabilities.

More importantly, people who go for electric vehicles are more likely to adopt such autonomous features rather than people driving ICE vehicles. I also am of the view that electric vehicle makers will be the first ones to launch autonomous driving features.

Software is seen as one of India's strengths, is there an opportunity for the hardware sector here?

That's a wonderful question, to be honest. I have a background in robotics and doing robotics in India is a difficult task because that is the juncture where